

Mohamed Awadalla

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EXPERIENCE

New York City Emergency Management

Jun 2025 – Present

Software Development & Automation Intern

Brooklyn, NY

- Built and operated a REST API event system supporting 1,000+ registrants with structured logging, failure tracking, and production monitoring
- Built a compliance tracking system with SQL database and automated reporting, serving 4 legal team members with reliable data pipelines
- Designed automated data pipelines with retry logic, dead-letter handling, and scheduled execution for production reliability
- Evaluated 8 platform vendors on technical capability, security, and integration fit; presented architectural recommendations to Chief Counsel

PROJECTS

LegalDocuMan | *Python, Flask, PostgreSQL, React, Redis, Docker, REST APIs*

[GitHub](#)

- Built a full-stack document-processing platform with a React frontend, Flask REST API, PostgreSQL persistence, and Redis-backed async workers for scalable background processing
- Designed RESTful endpoints with request validation, error handling, and comprehensive logging for production observability
- Deployed containerized services with Docker, managing API server, worker processes, Redis, and PostgreSQL in a multi-container architecture
- Implemented AI-powered document classification combining computer vision (RF-DETR, 94.9% precision) and OCR pipelines for automated data extraction

Custom Autograd Engine & Character-Level Language Model | *Python, PyTorch*

[GitHub](#)

- Implemented a reverse-mode autodiff engine from scratch with tensor operations, topological backpropagation, and finite-difference gradient checks validating against torch.autograd
- Trained a character-level neural language model with embeddings, MLP layers, and backpropagation on a 32,000-name dataset

Personal Portfolio | *React, TypeScript, Cloudflare*

moawadalla.com

- Built a React 18 + TypeScript SPA with custom CSS architecture, responsive layouts, and Intersection Observer-based scroll animations
- Built a Cloudflare Worker API with serverless deployment, CORS handling, and persistent data storage on Cloudflare D1

EDUCATION

Long Island University – Honors College

May 2026

Bachelor of Science in Computer Science

Brooklyn, NY

Honors: Dean's List, Dean Scholar

Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Computer Architecture, Machine Learning

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C++

Backend: Flask, FastAPI, Node.js, REST APIs, PostgreSQL, Redis, Docker

AI/ML: PyTorch, Autograd, Computer Vision, NLP, Model Fine-tuning, OCR

Infrastructure: Git, Linux, Docker, Docker Compose, Cloudflare Workers/Pages